2025 4th International Symposium on Computer Applications and Information Technology (ISCAIT) | 979-8-3315-4285-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/ISCAIT64916.2025.11010659

Formal Security Analysis and Innovative Application of Blockchain-based 5G AKA Protocol in Service Networks

Shi Guo
Jiangsu Automation Research Institute, Lianyungang, China
guo_shi@foxmail.com

***Abstract*—This paper proposes a blockchain architecture to address the challenges posed by the inadequate integration of technology and the absence of security analysis in the context of 5G communication and blockchain technology. The architecture integrates a blockchain-based 5G AKA protocol, a novel design that utilizes the tamper-proof characteristics of blockchain to enhance the security performance of the conventional protocol. The Tamarin-prover tool is applied to model and analyze the blockchain-based 5G AKA protocol, and the results indicate that the new protocol satisfies 7 of the 10 unsatisfied security authentication properties of the traditional 5G AKA protocol, improving security by 60%. This provides an effective strategy for enhancing 5G communication security.**

Keywords—Network security, Blockchain-based 5G AKA protocol, Formal Analysis, Blockchain-based service network architecture

I. INTRODUCTION

The rapid development of 5G networks, with features like high speed, low latency and large connectivity, enables its wide application in intelligent transportation, industrial automation and other fields, promoting digital transformation. Meanwhile, blockchain technology, as a decentralized distributed ledger, has been applied in finance, supply chain, government and other areas, showing strong capabilities in data security and trust establishment.

The fast-developing 5G networks with wide applications, combined with the secure and reliable blockchain technology, have been integrated in many fields. But current research and applications have issues: (1) Blockchain technology isn't fully integrated into 5G communication, lacking depth, and its security features aren't effectively used to enhance 5G core processes like authentication and key exchange. (2) There's a lack of analysis and proper improvement methods for the communication security of 5G-based service systems.

In addressing the limitations of existing research, this paper presents a novel approach to augmenting the service network architecture of the blockchain platform. The core component of this platform is the newly designed Blockchain-based 5G AKA protocol, which incorporates the blockchain's immutability to enhance the security of the traditional 5G AKA protocol. The paper then conducts a detailed and comprehensive analysis of the system's security attributes using a formal analysis tool. The contributions of this paper are as follows:

1. The paper proposes a service network architecture with the addition of a blockchain platform to address the identified weakness. This blockchain platform serves to record the critical messages of the service network communication in the form of transactions, thereby ensuring the security of the communication between the nodes of the service network. The tamper-proof capability of the blockchain platform is leveraged to guarantee the integrity of the network communications.

2. Regarding the service network architecture put forward in this study, an innovative Blockchain-based 5G AKA protocol is designed. It fully incorporates the security traits of blockchain, with the goal of enhancing the security and privacy of the authentication and key agreement procedures between user equipment and communication networks. This approach offers novel insights for future research on integrating 5G and blockchain, and can be extended to protocols like EAP-AKA.

3. In this paper, the proposed Blockchain-based 5G AKA protocol is modeled and analyzed for its security properties using the Tamarin-prover formal analysis tool. The formal analysis validation results indicate that among the 10 unsatisfied security authentication properties of the traditional 5G AKA protocol, 7 of them are satisfied in the improved Blockchain-based 5G AKA. Furthermore, the security of the improved protocol is significantly enhanced, reaching up to 60% compared with the original one.

II. RELATED WORKS

For the aspect of 5G communication fusion blockchain technology. On the one hand, the high-speed transmission and low-latency characteristics of 5G provide a better operating environment for the blockchain system, which helps to improve the transaction processing speed and system performance. Khan et al. study that in the field of supply chain management, the combination of 5G and blockchain technology can realize a more efficient and transparent supply chain management, and improve the overall efficiency [1][2]. Conversely, the decentralization, data integrity, and other characteristics inherent to blockchain can assist 5G in addressing challenges related to network security, data privacy, and other issues, thereby enhancing the network's trustworthiness and security. This combination has been demonstrated in practical applications in fields such as telemedicine and energy transactions [3][4][5]. However, there is a paucity of research that utilizes the blockchain platform to directly secure key parameters of the 5G communication process.

This research was funded by the Science and Technology Commission, CMC, Grant No. 23-TQ06-22-ZD-01-001-ZDJ-02.

For the formal analysis aspect of 5G AKA protocols. Basin et al. [6] proposed the first formal model of 5G AKA protocols, used Tamarin to formalize the 5G AKA protocols and extracted the security objectives from the 5G security standards. Cao et al. [7] designed a lightweight secure access authentication protocol to address the scenario of concurrent connections of massive devices in 5G networks and formally analyzed the protocol using two tools, ProVerif and Scyther. However, both tools have corresponding limitations. Liu et al. [8] formally analyzed the 5G AKA protocol based on 3GPP TS 33.501 v17.0.0 [9] using Tamarin and improved the protocol model to enhance the security properties of the protocol. This paper synthesizes the extant research results and methodologies in the domain of formal analysis. It employs the Tamarin to formally analyze the Blockchain-based 5G AKA protocol that is proposed in this paper.

III. Service Network Architecture

A. Original service network architecture

A service network that utilizes 5G communications generally employs the 5G AKA protocol to facilitate authentication and key exchange between the user and the operator's network. As illustrated in Fig. 1, this network architecture consists of three components: user equipment (UE), a serving network (SN) to which the subscriber is connected in close proximity, and a home network (HN) corresponding to the subscriber for the operator.

In this case, UEs communicate with the SN by means of a wireless link, which is insecure and vulnerable to attack. The communication between the SN and the HN, as well as within the HN, is through wired links, which can be considered secure.

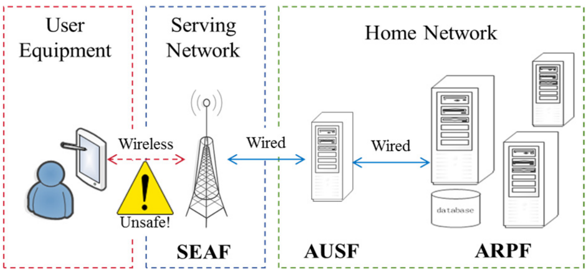

Fig. 1. Original service network architecture.

1) A UE, which is most commonly a smartphone or an IoT device, must be equipped with a 5G USIM card. The USIM card is a specialized type of hardware chip that stores the UE's SUPI (Subscription Permanent Identifier), SQN (Sequence Number), and K (Long-Term Symmetric Key) information.

2) The serving network (SN) is a pivotal component in the wireless communication infrastructure. The security anchor function module (SEAF) is a sub-module in the serving network that is responsible for forwarding messages between the UE of the user equipment and the HN of the attributing network. It is also responsible for completing the authentication and key exchange process for the UE and assisting in the negotiation process between the UE and the attributing network. Following the culmination of the aforementioned key exchange process, the anchor key K_{SEAF} is obtained, thereby ensuring the encryption protection service for the communication content between the UE and the SEAF.

3) Home network (HN): The authentication server function (AUSF) and the authentication credential repository and processing function (ARPF) are the key functional modules utilized in the home network by the 5G AKA protocol. The AUSF module is mainly responsible for authenticating the identity of the UE side and deriving the anchor key K_{SEAF}. The main function of the ARPF module is to store the privacy data involved in the protocol.

B. Blockchain-based service network architecture

As illustrated in Fig. 2, in contrast to the traditional service network architecture, the blockchain-based service network architecture establishes connections between the blockchain nodes via wired links on both the user equipment (UE) side and the service network side. This facilitates communication between the user equipment and the service network through the interconnectivity among the blockchain nodes.

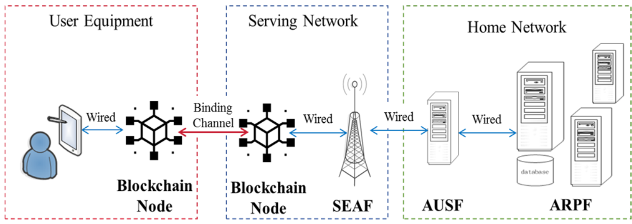

Fig. 2. Blockchain-based service network architecture.

1) Blockchain nodes: The blockchain platform is comprised of disparate blockchain nodes that can join the platform subsequent to authentication. These nodes record the information of the transmitted data between them and concurrently upload the confirmed data to the chain. The nodes execute the functions of privacy protection, tamper-proof, encrypted communication, and authority management to ensure the trustworthy transmission of the service network. The communication between blockchain nodes can be conceptualized as analogous to the use of bound channels for communication, which can be regarded as secure.

IV. Blockchain-Based 5G AKA Protocol Authentication Process, Formal Modeling And Verification Analysis

A. Blockchain-based 5G AKA protocol authentication process

The detailed authentication process of the Blockchain-based 5G AKA protocol is shown in Fig. 3.

The 5G AKA (Authentication and Key Agreement) protocol is a security protocol for authentication and key exchange between UEs and the network in 5G communication systems, which ensures the legitimacy of the two communicating parties through cryptographic algorithms and message exchanges and generates keys for secure communication. The Blockchain-based 5G AKA protocol innovatively proposed in this paper adds blockchain nodes based on the 5G AKA protocol in the 3GPP TS 33.501 v18.5.0 version [10] to further improve the overall authentication security.

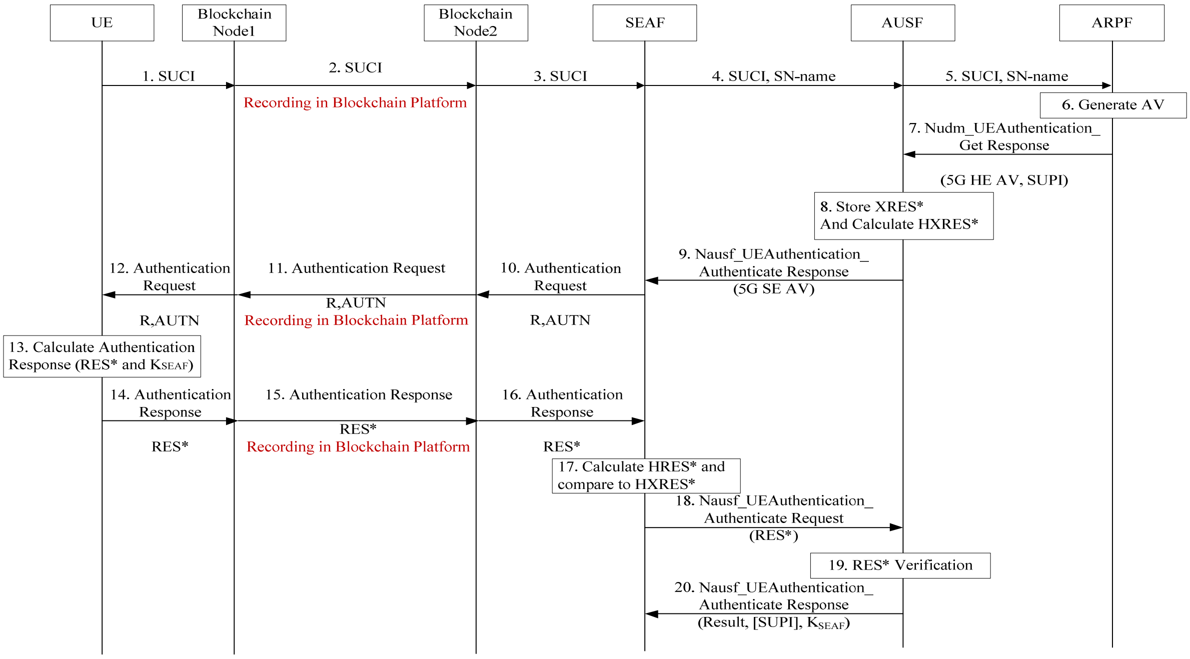

Fig. 3. Blockchain-based 5G AKA Protocol Authentication Process.

Step1: The UE encrypts the SUPI into a SUCI using the public key of the corresponding HN and sends the SUCI to BlockChain Node1.

Step2: Blockchain Node1 sends the SUCI directly to Blockchain Node2 in the SN network, and the related data is also recorded directly in the blockchain platform's data chain to ensure it cannot be tampered with.

Step3 and Step4: Blockchain Node2 sends the SUPI to SEAF, which determines the corresponding SN based on the SUPI and simultaneously sends the corresponding SN name to AUSF.

Step5: The AUSF will confirm the SN-name with the SN name, and after confirming the agreement, it will cache the SN-name and then send the SUCI and SN-name to the ARPF.

Step6 and Step7: The ARPF calculates the corresponding 5G HE AV according to SUCI and sends it back to AUSF. The 5G HE AV includes RAND, AUTN, XRES* and K_{AUSF}.

Step8: The AUSF stores the XRES* in the 5G HE AV and calculates its hash value HXRES* and computes the corresponding K_{SEAF} via the K_{AUSF}.

Step9: The AUSF shall then return the 5G SE AV (RAND, AUTN, HXRES*) to the SEAF.

Step10: SEAF receives the 5G SE AV and sends the corresponding RAND and AUTN parameters to Blockchain Node2 via the Authentication Request message.

Step11: Blockchain Node2 sends the RAND and AUTN to Blockchain Node1, and it is recorded on the blockchain platform for privacy and tamper resistance.

Step12: Blockchain Node1 sends RAND and AUTN to UE.

Step13: After receiving the RAND and AUTN information, the UE will verify the validity by the MAC and SQN in the AUTN. After passing the validation, the UE will calculate the RES* and the K_{SEAF}.

Step14: The UE sends the parameter RES* to the Blockchain Node1 via the Authentication Request message.

Step15: Blockchain Node1 sends the RES* to Blockchain Node2 and is recorded on the blockchain platform for privacy and tamper-proofing.

Step16: Blockchain Node2 sends the parameter RES* to SEAF.

Step17 and Step18: The SEAF calculates its hash value HRES* from the parameter RES*, then compares HRES* with HXRES*, and if it recognizes this authentication process from the SN's point of view, it sends RES* to the AUSF for the next verification step.

Step19: The AUSF receives the parameter RES* and checks if the parameters RES* and XRES* are identical. If they are identical, the AUSF recognizes the success of this authentication from the point of view of the HN.

Step20: When authentication is complete, the AUSF sends the keys K_{SEAF} and SUPI to the SEAF.

B. Blockchain-based 5G AKA protocol formal modeling

In this paper, we use Tamarin to formally model blockchain-based 5G AKA protocols, model security properties, and validate and analyze the related results. In this paper, Tamarin Prover 1.8.0 is used and deployed on Ubuntu 16.04 system.

A total of four participating entities are considered for formal modeling of the Blockchain-based 5G AKA protocol, which are the UE-Blockchain Node1 entity, the Blockchain Node2-SEAF entity, the AUSF entity and the ARPF entity.

Among them, since UE-Blockchain Node1 uses a wired secure channel, the messages transmitted over the secure channel cannot be eavesdropped and tampered by an attacker, so the two can be viewed as a unified entity binding modeling in modeling. Blockchain Node2-SEAF entity is consistent with the above.

In addition, the communication between Blockchain Node1 and Blockchain Node2 works through the blockchain platform for critical data upload, which ensures data privacy and tamper-proof, so this paper models the modeling between UE-Blockchain Node1 entity and Blockchain Node2-SEAF entity using the binding channel way to model the capabilities of the blockchain platform.

C. Blockchain-based 5G AKA protocol security property modeling

In terms of security attributes, this paper compares with the conventional 5G-AKA protocol in terms of protocol executability, secrecy, and agreement.

1) Executability: Ensure the protocol is modeled so the process can be executed with the primitives to ensure the protocol's executability.

```
lemma trace_exists:
  exists-trace
  "
  Ex #i. HN_END()@i
  & not (Ex X #r. Rev(X)@r)
  & (All AUSF AUSF2 #j #k. StartAUSFSession(AUSF)@j &
        StartAUSFSession(AUSF2)@k ==> #j = #k)
  & (All SN1 SN2 #j #k. StartSeafSession(SN1)@j &
        StartSeafSession(SN2)@k ==> #j = #k)
  & (All ARPF1 ARPF2 #j #k. StartARPFSession(ARPF1)@j &
        StartARPFSession(ARPF2)@k ==> #j = #k)
  "
```

2) Secrecy: Since the blockchain-based 5G AKA protocol proposed in this paper mainly enhances the communication channel between the UE and SEAF, the secrecy of the K_{SEAF} is mainly modeled from the perspective of the UE, SEAF.

The secrecy property of the K_{SEAF} is modeled from the UE perspective as follows:

```
lemma secrecy_UE:
  "
  All a b c d t #i. Secret1(<a,b,c,d>,<'UE', a>, t) @i
  & not(Ex SUPI ARPF #r. RevealKforSUPI(SUPI)@r)
  ==> not (Ex #j. K(t)@j)
  "
```

The secrecy property of the K_{SEAF} is modeled from the SEAF perspective as follows:

```
lemma secrecy_SEAF:
  "
  All a b c d t #i . Secret1(<a,b,c,d>,<'SEAF', b>, t) @i
  & not(Ex SUPI ARPF #r. RevealKforSUPI(SUPI)@r)
  ==> not (Ex #j. K(t)@j)
  "
```

3) Agreement: Since the Blockchain-based 5G AKA protocol proposed in this paper mainly enhances the communication channel between UEs and SEAFs, the agreement of the parameters SUPI, SN-name and the key K_{SEAF} is mainly modeled from the UE perspective.

From the UE's point of view, non-injective agreement over SUPI and SN-name, and non-injective agreement and injective agreement over the key K_{SEAF} are considered between it and SEAF, AUSF, and ARPF, respectively.

Non-injective agreement [11] means that multiple different running instances of the protocol can produce the same output or achieve the same authentication result. Injective agreement [11] means that each different instance of the protocol running in an authentication protocol produces a unique output or authentication result.

D. Analysis of validation results

1) Analysis of the validation of executability and secrecy

It is experimentally verified that the Blockchain-based 5G AKA protocol proposed in this paper satisfies the executability.

The validation results for the secrecy satisfaction are shown in Table 1, which compares the traditional 5G AKA protocol with the Blockchain-based 5G AKA protocol proposed in this paper, where√represents satisfied and × represents unsatisfied.

TABLE I. ANALYSIS OF THE VALIDATION OF SECRECY

Protocol	Participant	K_{SEAF} Secrecy
Traditional 5G AKA	UE	√
	SEAF	√
Blockchain-based 5G AKA	UE	√
	SEAF	√

Table 1 shows that the proposed blockchain-based 5G AKA protocol in this paper satisfies the secrecy properties of the key K_{SEAF} from the perspective of UE, SEAF, which is consistent with the secrecy results of the traditional 5G AKA protocol.

TABLE II. ANALYSIS OF THE VALIDATION OF AGREEMENT

Protocol	Participant	SUPI Non-injective agreement	SN-name Non-injective agreement	K_{SEAF} Secrecy Non-injective agreement	K_{SEAF} Secrecy Injective agreement
Traditional 5G AKA	UE→ SEAF	×	×	×	×
	UE→ AUSF	√	×	×	×
	UE→ ARPF	√	×	×	×
Blockchain-based 5G AKA	UE→ SEAF	×	√	×	×
	UE→ AUSF	√	√	√	√
	UE→ ARPF	√	√	√	√

2) Analysis of the validation of agreement

The verification results for the satisfaction of the agreement for authentication are shown in Table 2, which compares the traditional 5G AKA agreement and the Blockchain-based 5G AKA agreement proposed in this paper, where √ stands for satisfied and × stands for unsatisfied.

Table 2 shows that the Blockchain-based 5G AKA protocol proposed in this paper outperforms the traditional 5G AKA protocol in terms of SUPI non-injective agreement, SN-name non-injective agreement, key K_{SEAF} non-injective agreement, and injective agreement.

For the 10 unsatisfied security authentication properties of the traditional 5G AKA protocol, 7 of the 10 unsatisfied security authentication properties of the improved Blockchain-based 5G AKA are satisfied, and the security of the improved protocol is improved by 60% compared to the original one.

V. CONCLUSION

Current research has the weaknesses that blockchain technology has not been deeply integrated into the 5G communication process and its security features have not been effectively utilized to enhance the reliability of core processes. To address these issues, this paper proposes a business network architecture with an added blockchain platform and innovatively designs the Blockchain-based 5G AKA protocol for this architecture. By using the Tamarin-prover formal analysis tool to model and analyze the security attributes of the protocol, it is strongly verified that the proposed Blockchain-based 5G AKA protocol has achieved a significant improvement of up to 60% in security properties compared to the traditional 5G AKA protocol. This provides a practical solution and theoretical support for enhancing the security of 5G communication.

Future work will consider two aspects. One is to put the business network architecture with an added blockchain platform into practical application. The other is to explore more possibilities for the organic combination of 5G network protocols and blockchain technology.

ACKNOWLEDGMENT

This research was funded by the Science and Technology Commission, CMC, Grant No. 23-TQ06-22-ZD-01-001-ZDJ-02.

REFERENCES

[1] Khan, Syed Abdul Rehman , and Z. Yu . "A Systematic Literature Review: Blockchain Technology and Organizational Theories in the Perspective of Supply Chain Management." Journal of Physics: Conference Series 1910.1(2021):012011 (6pp).

[2] Agi, M. A. N. , and A. K. Jha . "Blockchain technology in the supply chain: An integrated theoretical perspective of organizational adoption." International journal of production economics May(2022):247.

[3] Fan, Kai , et al. "Blockchain-based efficient privacy preserving and data sharing scheme of content-centric network in 5G." IET Communications 12.5(2018):527-532.

[4] An, Sung Yong , G. Ngayo , and S. P. Hong . "Enhancing 5G Antenna Manufacturing Efficiency and Reliability through Blockchain and Smart Contract Integration: A Comprehensive AHP Analysis." Applied Sciences (2076-3417) 14.6(2024).

[5] Islam, Anik , S. Chae , and S. Y. Shin . "Social Internet of Things (SIoT) and Blockchain: Research opportunities and challenges." Proceedings of Symposium of the Korean Institute of communications and Information Sciences 2018.

[6] Basin D , Dreier J , Hirschi L ,et al. "A Formal Analysis of 5G Authentication[J]." ACM, 2018.DOI:10.1145/3243734.3243846.

[7] Park, Kisung , and Y. Park . "On the Security of a Lightweight and Secure Access Authentication Scheme for Both UE and mMTC Devices in 5G Networks." Applied Sciences (2076-3417) 12.9(2022).

[8] LIU D, WANG Z Y, LI D W, GUAN Z Y, SUN Y, LIU J W. "Formal analysis and improvement methods of 5G AKA protocol based on tamarin[J]." Journal of Cryptologic Research, 2022, 9(2): 237–247. [DOI: 10.13868/j.cnki.jcr.000515]

[9] 3GPP. "Security Architecture and Procedures for 5G System[S]." 3GPP TS 33.501, v17.0.0. 2020.

[10] 3GPP. "Security Architecture and Procedures for 5G System[S]." 3GPP TS 33.501, v18.5.0. 2024.

[11] Tanarin team. "Tamarin-Prover Manual:Security Protocol Analysis in the Symbolic Model". 2023